



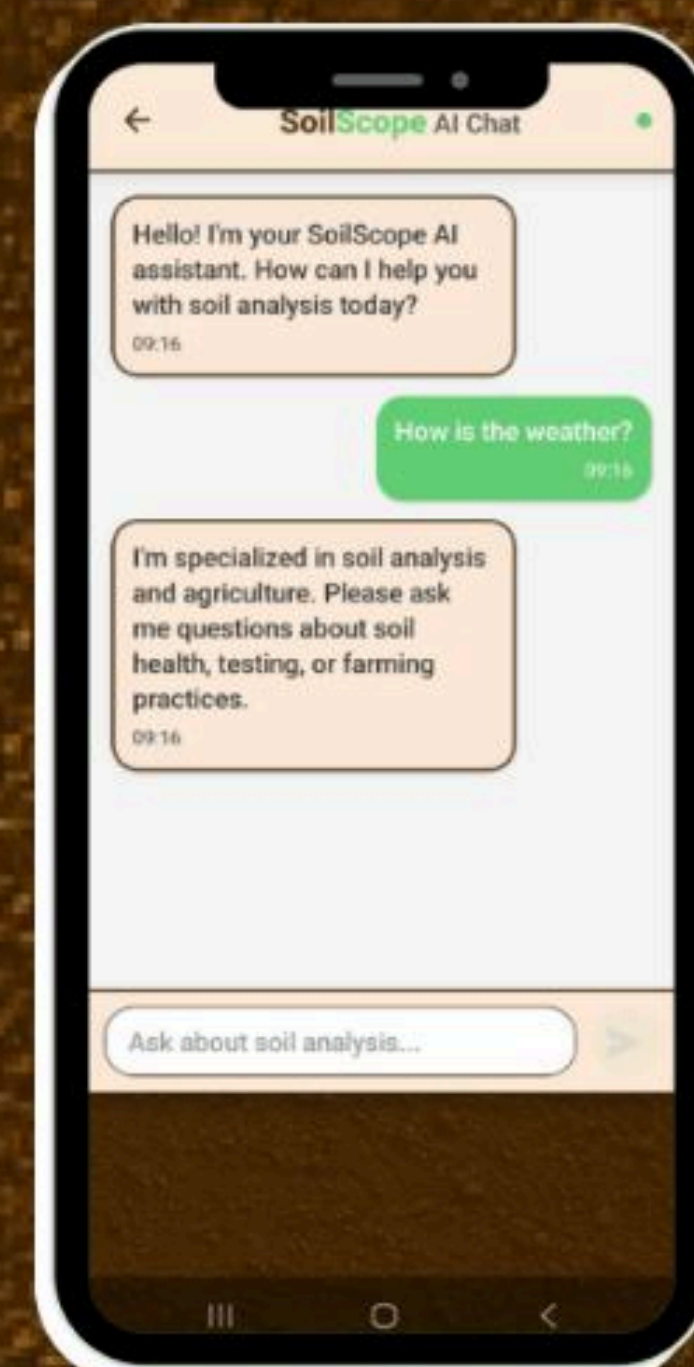
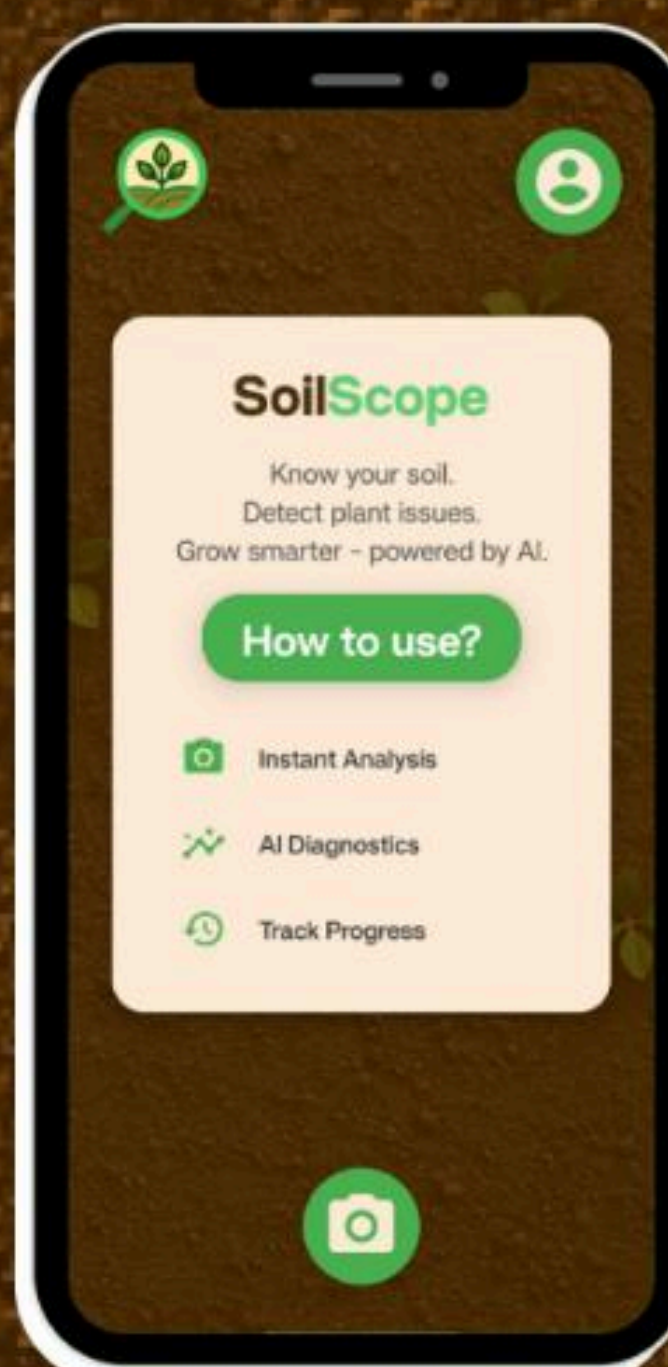
SOILSCOPE

An AI-Powered Mobile Application for Soil Analysis and Disease Detection



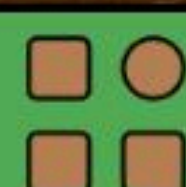
ABSTRACT

A mobile application leveraging AI and computer vision to analyze soil quality and detect plant diseases. SoilScope integrates offline functionality, engagement, and chatbot support to empower small-scale farmers and gardeners with accessible, science-based recommendations.



OBJECTIVES

- Design and develop SoilScope for real-time soil analysis and disease detection.
- Provide sustainable farming recommendations tailored to local contexts.
- Evaluate usability and effectiveness using SUS and ISO/IEC 25010 framework.



FEATURES

Real-Time Soil Analysis

AI-powered image recognition for soil quality and soil health diagnostics.

Smart Gardening Tips

Recommendations on watering, sunlight, and organic care practices.

Chatbot Support

Interactive guidance for farmers and gardeners, available on-demand.



METHODOLOGY

Adopted Scrum Agile Framework with iterative sprints.

Tools: React Native, Firebase, TensorFlowJS, Expo Camera

Evaluation: SUS survey + ISO/IEC 25010 quality framework

Phase 1 : Planning

Defined scope, user needs, consulted expert (Mr. Zabala).

Phase 2 : Design

Wireframes, UI/UX focused on accessibility, engagement.

Phase 3 : Development

React Native, Firebase, TensorFlowJS; CNN models for soil & disease detection.

Phase 4 : Testing

SUS surveys, ISO/IEC 25010 evaluation with end-users & IT experts.

Phase 5 : Deployment

Android release, offline functionality, iterative sprints.

Phase 6 : Maintenance

Continuous refinement, planned dataset expansion & weather API.



REFERENCES

Köse et al. (2024) :

AI in Smart Agriculture

Da Silva et al. (2022) :

Smartphone-based soil diagnostics

Bautista & Ramos (2020) :

Digital innovations in Philippine agriculture

Schwaber & Sutherland (2020) :

Scrum framework



RESULTS

High usability scores (SUS), strong reliability, effective soil and disease diagnostics. SoilScope demonstrates potential as a cost-efficient, farmer-friendly digital agriculture tool aligned with SDGs and the Philippine Development Plan.



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